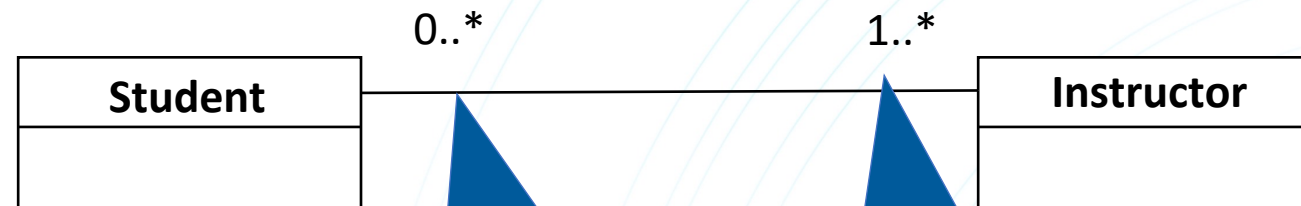


- If two classes in a model need to communicate with each other, there must be a link between them, and that can be represented by an association (connector).



One instructor can teach 0 or more students.

One student learns from 1 or more instructors.

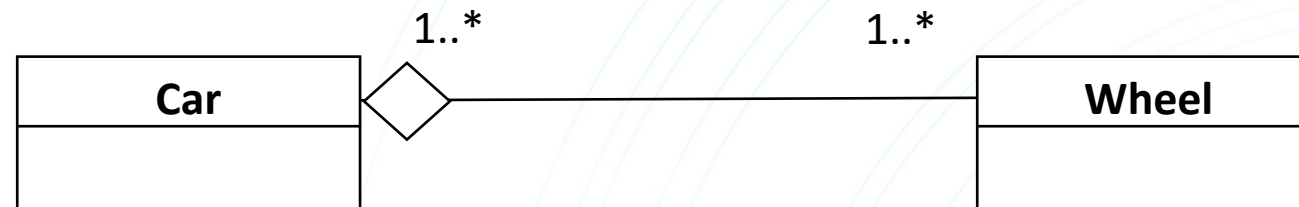


**Multiplicities** indicate how many objects take part in an association.

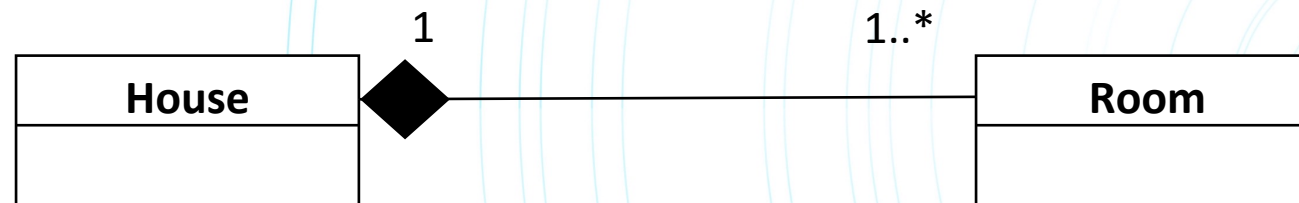


The unified modeling language (**UML**) is a graphical language to describe OO models.

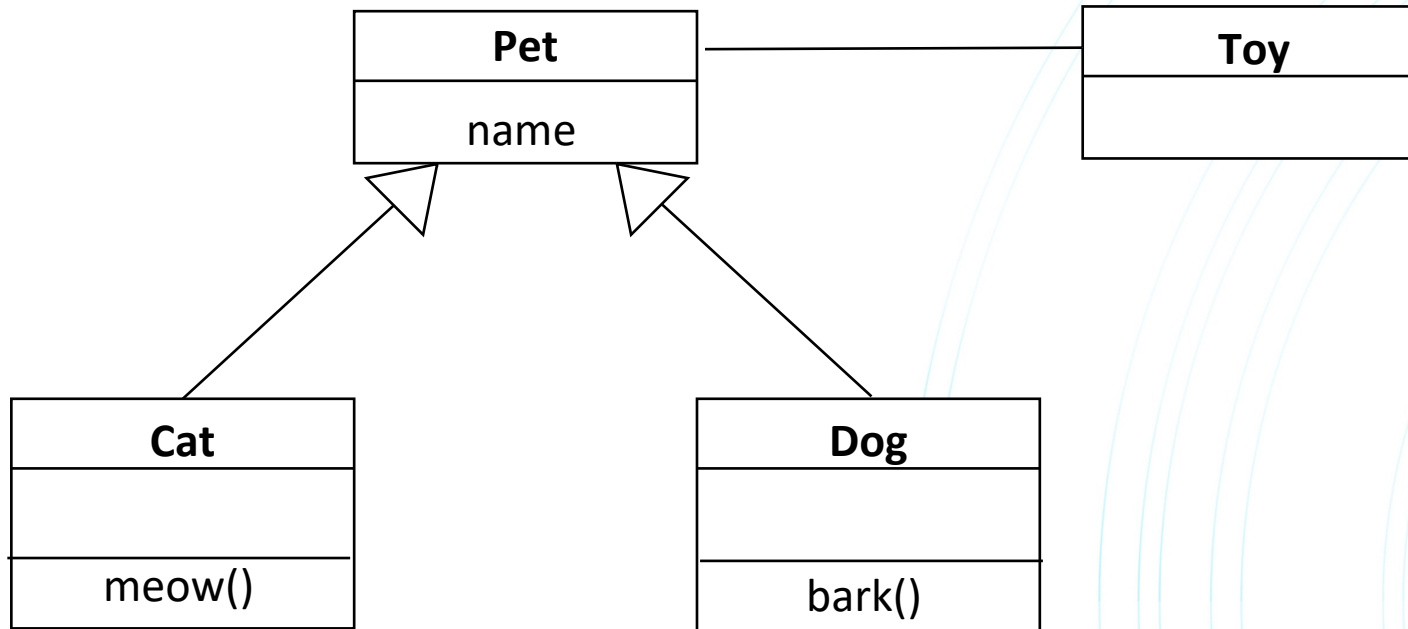
- Describe a **has-a** relation (in the sense of „consists of“)
- In an **aggregation**, the elements can exist independently from the composite



- In a **composition**, the elements **cannot exist** independently from the composite



- Describes an **is-a** relationship



- A cat **is a** pet
- A dog **is a** pet
- All pets have a name and can have toys
- But only cats can meow and dogs can bark

# Task 1 - Factory in UML

